

Shabab Ahmed

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PROFESSIONAL PROFILE

Ph.D. Candidate in Economics specializing in contract theory and structural modeling. Applied the latter during a pricing-focused economics internship at Amazon.

EDUCATION

Ph.D. in Economics, University of Washington

Expected Spring 2027

Dissertation:

- [Spend Now or Later? Dynamic Moral Hazard with a Fixed Budget](#) (JMP). Models how a manager decides when to spend a fixed budget. The manager holds spending until late in the cycle, which resembles year-end spending spikes even without use-it-or-lose-it rules. Headquarters would spend early instead, so delegation cuts its profit but can raise total welfare.
- [High-Dimensional Instrumental Variables Estimation of the Almost Ideal Demand System](#). Applies two-stage Lasso IV with a debiasing correction for shrinkage bias to estimate own- and cross-price elasticities with hundreds of collinear prices and many instruments. In Monte Carlo simulations, estimates stay stable as the number of goods nears the sample size, where OLS and 2SLS break down. Currently extending this estimator to merger simulation, testing whether a high-dimensional AIDS predicts different price effects than multi-stage AIDS or BLP.
- [Tanking from the Start: Dynamic Moral Hazard with Change in Preference](#). Studies how anticipated preference changes affect incentives today, applied to the NFL draft policy. The model predicts that teams with weaker playoff prospects ease off from the start of the season, not only after elimination. Currently testing this prediction with quasi-experimental methods.

M.A. in Economics, University of Washington

Dec 2022 · GPA 3.89

M.S. in Applied Mathematics, University of Washington

Jun 2020 · GPA 3.93

B.A. in Mathematics and Economics, Colby College

May 2019 · GPA 3.93

Magna Cum Laude, Phi Beta Kappa, Distinction in Economics & Mathematics, Dean's List (6 semesters).

EXPERIENCE

Economist Intern, Amazon, Bellevue, WA

Jul 2025 – Oct 2025

- Estimated a discrete choice model for pre-launch Project Kuiper from large-scale survey data, using reduced-form regressions to choose the model's consumer characteristics and interactions. Calibrated the estimates by matching external benchmarks and correcting stated willingness to pay for hypothetical bias using published estimates.
- Built a Python tool that projected market share and revenue as a calibrated range, which the team used to compare counterfactual pricing and product configurations. Presented a technical and business report to senior leadership across multiple teams, who evaluated it as an input to pricing decisions.

Graduate Teaching Assistant & Independent Instructor, University of Washington

2019 – Present

- Independent Instructor for Game Theory and Managerial Economics, with full responsibility for course delivery, assignments, and assessment.
- Teaching Assistant for Graduate Microeconomic Theory and undergraduate courses, leading discussion sections and review sessions.

EARLIER PROJECTS

[NBA Decade Comparison: PCA and Machine Learning Analysis](#)

Winter 2020

- Partnered with a classmate to build classifiers on 13,000 player-season observations (~95 features scraped from Basketball Reference). Used PCA and LDA to predict player position and decade. Used L1-regularized logistic regression to predict Win Shares quartile.
- Achieved 72.6% accuracy predicting Win Shares quartile for 1990s-decade players, cross-validated over 50 iterations.

[Game Theory in Cricket](#)

Spring 2019

- Built a game-theoretic model of bowler-batsman strategy and tested Nash equilibrium predictions against a hand-coded dataset of bowler-batsman interactions from 2017 ICC Champions Trophy match commentary.
- Found predicted strategy frequencies matched actual play within one bootstrapped standard deviation under a simplified model, though the result did not hold under more granular or alternate specifications.

TECHNICAL SKILLS

Econometrics & Causal Inference: IV, DiD/event study, linear and logistic regression, bootstrapping, Monte Carlo simulation

Structural Modeling: demand estimation (AIDS, discrete choice), dynamic optimization

Machine Learning: Lasso/Elastic Net, debiased ML, PCA, LDA, cross-validation

Languages & Tools: Python, R, Stata, SQL, MATLAB, Sage, Mathematica, LaTeX, Qualtrics, Excel

Languages: English (fluent), Bengali (fluent), Hindi (conversational)